

Self-Interaction-Free Levy–Perdew–Sahni Density Functional Theory

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I. INTRODUCTION

Levi, Perdew, and Sahni derived a Schrödinger-like equation for a square-root of the electron density

$$\left\{ -\frac{1}{2}\nabla^2 + v_{\text{eff}}([n]; \mathbf{r}) \right\} n^{1/2}(\mathbf{r}) = \mu n^{1/2}(\mathbf{r}), \quad (1)$$

where $v_{\text{eff}}([n]; \mathbf{r})$ acts as the multiplicative effective potential. The eigenvalue μ represents the chemical potential, which corresponds to the negative of the ionization energy in exact DFT [1]. The exact form of the effective potential has to be approximated

$$v_{\text{eff}}([n]; \mathbf{r}) = v_{\text{ext}}(\mathbf{r}) + \int \frac{n(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r}' + \frac{\delta G[n]}{\delta n(\mathbf{r})}. \quad (2)$$

Recently, we have shown that the Fermi–Amaldi functional is the exact exchange in LPS DFT. Therefore, the LPS energy functional can be recasted into the sum of exact terms and an approximate functional $G[n]$, where exact terms serve as a ‘bosonic baseline’ for the total energy.

Acharya *et al.* [2] as well as Cedillo *et al.* [3] explored the possibility for including the N -dependent factor into the approximate functionals to improve their performance. Here, we elaborate on this idea through analytical derivation of N -dependent self-interaction-corrected $G[n]$ functionals in LPS theory. Our results suggest a universal way of developing such approximations without empirical fitting and with straightforward generalization to spin-polarized systems.

In one-determinantal approximations (Hartree–Fock, Kohn–Sham DFT), the exact exchange, $K[n]$, is given by

$$K[\{\varphi_i\}] = \sum_{i,j}^N K_{ij} = \sum_{i=1}^N K_{ii} + \sum_{i \neq j}^N K_{ij}, \quad (3)$$

where the sum over i only represents the self-exchange and the sum over all distinct pairs of i and j represents the interelectronic exchange contribution. Each exchange integral, K_{ij} , is expressed via spin-orbitals as

$$K_{ij} = \int \int \frac{\varphi_i^*(\mathbf{x}) \varphi_j(\mathbf{x}) \varphi_j^*(\mathbf{x}') \varphi_i(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} d\mathbf{x} d\mathbf{x}'. \quad (4)$$

II. SELF-INTERACTION-CORRECTED LPS THEORY

In electronic structure theory, an exchange functional serves two physical roles: the interelectronic exchange

and the self-interaction. As we showed, the FA functional is the exact exchange in LPS DFT. However, due to the nature of the LPS reference system, the role of the exact exchange (FA) in LPS is strictly reduced to the cancellation of self-repulsion between coulombically-interacting particles. This holds true for any N since, within LPS theory, they all occupy the same orbital which is the condition when the Fermi–Amaldi functional becomes exact and N -electron self-interaction-free [4]. Thus, the role of the interelectronic exchange is played by an approximate exchange functional, e.g., LDA or GGA approximations.

Approximate functionals can be shown to contain a spurious approximation to the self-interaction part, which is exactly captured by FA. Therefore, we propose to describe an exchange part in LPS theory by a hybrid of the Fermi–Amaldi term and a fraction of the approximate exchange excluding its self-interaction part.

III. DISCUSSION

A. Spin-Dependence of Exact and Approximate Functionals

Any approximate or exact functional can be derived for particles of arbitrary spin by considering the spin-statistics of the system. An important example is the exact exchange functional expressed in terms of one-particle reduced density matrix (1-RDM) $\rho(\mathbf{r}; \mathbf{r}') = \sum_{i=1}^{N/M} \phi_i^*(\mathbf{r}) \phi_i(\mathbf{r}')$, where $\phi_i(\mathbf{r})$ is a spatial orbital, as

$$E_{\text{x}}^{\text{exact}}[\rho; M] = -\frac{1}{2M} \iint \frac{|\rho(\mathbf{r}; \mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'. \quad (5)$$

Here, the explicit spin-dependence of the functional is consequential to the spin-dependence of the wave function from which the 1-RDM is derived. For fermions with $M = N$, Eq. (5) reduces to the Fermi–Amaldi (FA) exchange functional [5]

$$E_{\text{x}}^{\text{FA}}[\rho] = -\frac{1}{2N} \iint \frac{\rho(\mathbf{r}) \rho(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d\mathbf{r} d\mathbf{r}'. \quad (6)$$

Another similar example is the non-interacting kinetic energy functional

$$T_{\text{s}}[\rho; M] = \frac{M}{2} \sum_{i=1}^{N/M} \int |\nabla \phi_i(\mathbf{r})|^2 d\mathbf{r}, \quad (7)$$

which reduces to the von Weizsäcker (vW) kinetic energy functional for fermions with $M = N$ [6, 7]

$$T_{\text{vW}}[\rho] = \frac{1}{8} \int \frac{|\nabla \rho(\mathbf{r})|^2}{\rho(\mathbf{r})} d\mathbf{r}. \quad (8)$$

The Fermi–Amaldi functional is the exact exchange for one-electron systems and equivalent to Eq. (5) for two-electron singlets [8, 9]. This parallels the von Weizsäcker kinetic energy functional (8), which is exact in these same limits but fails for many-electron systems [6, 7].

The s -dependence of approximate functionals, such as LDA exchange or Thomas–Fermi kinetic energy, can be shown to arise from their statistical derivations [4, 7]

$$T_{\text{TF}}[\rho; M] = \left(\frac{2}{M}\right)^{2/3} C_{\text{F}} \int \rho^{5/3}(\mathbf{r}) d\mathbf{r}, \quad (9)$$

$$E_{\text{x}}^{\text{LDA}}[\rho; M] = \left(\frac{2}{M}\right)^{1/3} C_{\text{x}} \int \rho^{4/3}(\mathbf{r}) d\mathbf{r}, \quad (10)$$

leading to explicit dependence on electron number for fermions with $M = N$.

B. Explicit Dependence on Electron Number of Approximate Functionals

Consider an orbital-free energy functional

$$E[n] = T_{\text{W}}[n] + T_{\text{P}}[n] + V_{\text{ext}}[n] + J[n] + E_{\text{xc}}[n], \quad (11)$$

where $T_{\text{P}}[n]$ and $E_{\text{xc}}[n]$ must be approximated. For one-electron systems, $T_{\text{P}}[n]$ must vanish and $E_{\text{xc}}[n]$ must reduce to E_{x}^{FA} to recover the exact energy. For a two-electron singlet the same requirements are restoring the HF energy. For higher electron numbers, these constraints no longer apply, but it is still desirable that approximate functionals reduce to the correct limits for small N .

One possible approximation for the exchange energy is the spin-polarized LDA exchange functional

$$E_{\text{x}}^{\text{LDA}}[n_{\alpha}, n_{\beta}] = 2^{1/3} C_{\text{x}} \int \left[n_{\alpha}^{4/3}(\mathbf{r}) + n_{\beta}^{4/3}(\mathbf{r}) \right] d\mathbf{r}, \quad (12)$$

where $C_{\text{x}} = -3/4(3/\pi)^{1/3}$. This exchange functional is known to depend on the spin-statistics of the particles it is describing through its prefactor

$$E_{\text{x}}^{\text{LDA}}[n; M] = \left(\frac{2}{M}\right)^{1/3} C_{\text{x}} \int n(\mathbf{r})^{4/3} d\mathbf{r}, \quad (13)$$

where $M = 2s + 1$ and s is the spin quantum number. For boson-like fermions with $M = N$, the prefactor becomes $(2/N)^{1/3} C_{\text{x}}$. Subtracting this from the original LDA exchange functional yields the N -dependent correction

$$E_{\text{x}}^{\text{LDA}}[n_{\sigma}; N_{\sigma}] = 2^{1/3} \left(1 - \frac{1}{N_{\sigma}^{1/3}}\right) C_{\text{x}} \int n_{\sigma}(\mathbf{r})^{4/3} d\mathbf{r}, \quad (14)$$

where n_{σ} is the spin density for spin component σ and the subtracted term is treated as a self-interaction contribution. This correction vanishes in the large- N limit and cancels out the LSDA exchange energy completely for one- and two-electron singlets, yet for two-electron triplets this is not true.

C. Generalized Spin-Scaling Relations for Exact and Approximate Functionals

Unlike statistical approximations such as Thomas–Fermi or Slater exchange, both FA and vW maintain their functional forms for particles of arbitrary spin [4]. Applying spin-scaling relations [4, 10], the spin-polarized FA and vW functionals are expressed as sums over the spin-density components

$$E_{\text{x}}^{\text{FA}}[\rho_{\alpha}, \rho_{\beta}, \dots] = \sum_{\sigma=1}^M E_{\text{x}}^{\text{FA}}[\rho_{\sigma}] \quad (15)$$

$$T_{\text{vW}}[\rho_{\alpha}, \rho_{\beta}, \dots] = \sum_{\sigma=1}^M T_{\text{vW}}[\rho_{\sigma}], \quad (16)$$

these expressions represent the spin-resolved generalizations of Eq. (6) and Eq. (??), respectively. In contrast, statistical approximations scale with the spin degeneracy M according to

$$E_{\text{x}}^{\text{LDA}}[\rho_{\alpha}, \rho_{\beta}, \dots] = M^{1/3} \sum_{\sigma=1}^M E_{\text{x}}^{\text{LDA}}[\rho_{\sigma}] \quad (17)$$

$$T_{\text{TF}}[\rho_{\alpha}, \rho_{\beta}, \dots] = M^{2/3} \sum_{\sigma=1}^M T_{\text{TF}}[\rho_{\sigma}]. \quad (18)$$

This scaling behavior applies to most approximations on the DFT Jacob’s ladder, where spin dependence enters either via explicit prefactors (as in Slater exchange or Thomas–Fermi) or through variables such as the Fermi wave vector, k_{F} [4, 7].

Following Finzel *et al.* [11], we partition the total orbital-free energy (11) into bosonic (E_{B}) and fermionic (E_{F}) components. Bosonic functionals can be derived by substituting the normalized square root of the electron density into the corresponding orbital-dependent expressions. This yields the decomposition $E[n] = E_{\text{B}}[n] + E_{\text{F}}[n]$, where $E_{\text{B}}[n] = T_{\text{vW}}[n] + V_{\text{ext}}[n] + U[n] + E_{\text{x}}^{\text{FA}}$ and $E_{\text{F}}[n] \equiv G[n]$. The bosonic part comes out of the derivation, while the fermionic part includes approximate corrections to the boson-like kinetic and particle-particle interaction parts.

This decomposition suggests that approximations to the functional $G[n]$ are to be developed as post-boson-like energy terms, without necessarily splitting them into kinetic and exchange-correlation contributions. Consequently, while the Fermi–Amaldi functional is the exact orbital-free exchange, it constitutes only the part of the interacting electrons exchange. Improvements of this aspect are under study.

D. Self-Interaction Corrected Energy Functionals

Consider the energy functional of Eq. (11) for one-electron systems. In this case, the approximate functional $G[n]$ must reduce to the FA functional to restore

the exact energy. For two-electron singlets, the HF energy is exactly recovered when $G[n] = E_x^{\text{FA}}[n]$. Thus, the ‘bosonic’ contribution is treated exactly by the sum of the Weizsäcker, classical Coulomb and the Fermi–Amaldi functionals since they are the expectation values of the corresponding operators. For systems with more than two electrons, both the Pauli and the KS exchange–correlation functionals are required to restore the exact energy.

Conversely, when an approximate functional yields a non-zero value for one-electron systems or two-electron singlets, it introduces self-interaction error. This error can be eliminated by subtracting the ‘bosonic’ contribution, which is derived from the functional’s dependence on the spin statistics of the reference system [4]. Specifically, for fermions with spin degeneracy $M = N$, the explicit M -dependence of the functional becomes a dependence on the total particle number N .

As an example, consider the Dirac exchange functional (LDA) (??). Its spin-dependent form for fermions with degeneracy M is obtained by scaling the standard electronic functional of Eq. (??) by a factor of $(2/M)^{1/3}$ [4]. The bosonic limit corresponds to the case where $M = N$. Consequently, the self-interaction corrected (SIC) spin-polarized [4, 10] LDA exchange functional is given by

$$E_x^{\text{LDA-SIC}}[n] = 2^{1/3} \sum_{\sigma=\alpha,\beta} \left(1 - \frac{1}{N_\sigma^{1/3}}\right) E_x^{\text{LDA}}[n_\sigma], \quad (19)$$

where n_σ is the spin density for spin component σ . This functional vanishes for one-electron systems and two-electron singlets, yet correctly yields a non-zero value for two-electron triplets.

TABLE I. SCF atomic energies (Hartree) within the 31 s -type basis set. MAEs and rMAEs are given with respect to the UHF/UGBS.

Atom	Eq. (21)	Eq. (22)	UHF
H	−0.5000	−0.5000	−0.5000
He	−2.8617	−2.8617	−2.8617
Li	−7.0356	−8.5470	−7.4328
Be	−13.2319	−19.0194	−14.5730
B	−22.1259	−35.7426	−24.5293
C	−33.9335	−60.1803	−37.6900
N	−48.7991	−93.7963	−54.4045
O	−65.9537	−138.0543	−74.8140
F	−86.4703	−194.4182	−99.4114
Ne	−110.6741	−264.3516	−128.5471
MAE	5.32	37.27	–
rMAE	8.34	50.90	–

The same logic can be applied to any approximate Pauli or exchange–correlation functional to eliminate self-interaction errors in one-electron systems and two-electron singlets. For instance, the Thomas–Fermi kinetic energy functional (??) can be corrected as

$$T_P^{\text{TF-SIC}}[n] = 2^{2/3} \sum_{\sigma=\alpha,\beta} \left(1 - \frac{1}{N_\sigma^{2/3}}\right) T_{\text{TF}}[n_\sigma]. \quad (20)$$

The corrections of this type vanish in large- N limit restoring the original approximate functional. The resulting functionals (19) and (20) are resembling the functionals of [2, 3] that were obtained by fitting the corresponding approximate functionals to the post-SCF atomic energies.

To implement the LDA-SIC and TF-SIC approximations for the exchange–correlation and Pauli terms, respectively, we solved Eq. (??) using the functional derivatives of Eqs. (19) and (20), treating N as a parameter [12, 13]. Consequently, the resulting potentials are simply scaled versions of the standard LDA and TF potentials. Table III reports the SCF results for H–Ne, comparing the Eq. (11) with

$$G[n] = E_x^{\text{FA}}[n] + T_P^{\text{TF-SIC}}[n] + E_x^{\text{LDA-SIC}}[n] \quad (21)$$

to the pure ‘bosonic’ model with

$$G[n] = E_x^{\text{FA}}[n]. \quad (22)$$

All calculations employed the 31 s -type basis set and were compared against UHF/UGBS reference values.

Although the proposed TFD-SIC local functional does not surpass the accuracy of the optimal- λ results in Table I, it exhibits superior convergence properties. Specifically, it permits convergence to tighter thresholds, e.g., 1.0E-8.

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